



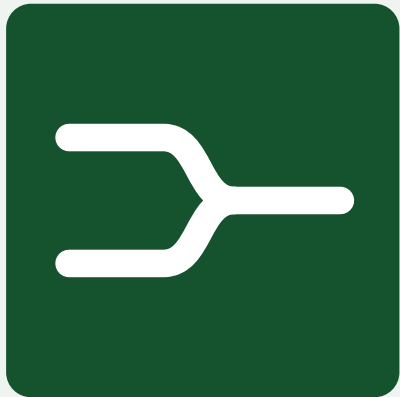
ReconMed

Automated Medicine Reconciliation

Reconciled before the visit.

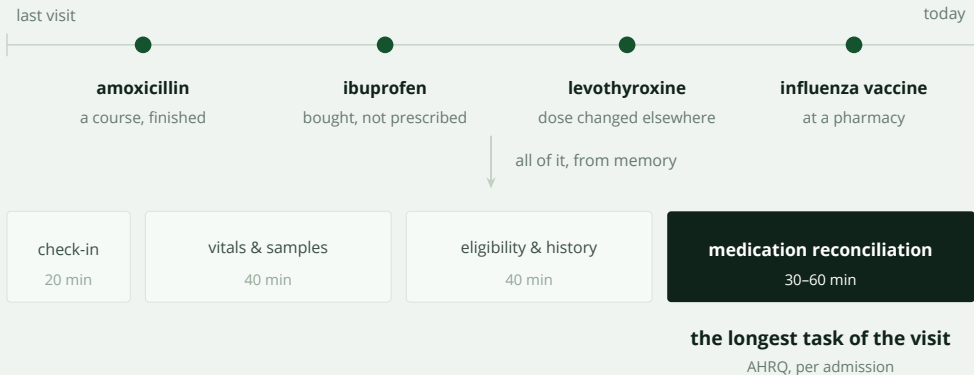
Pre-visit medication reconciliation
for clinical trial sites.

HackMIT 2026



The medication list is stale, and the visit is now

Every medication a participant takes besides the study drug is collected once, at the visit.



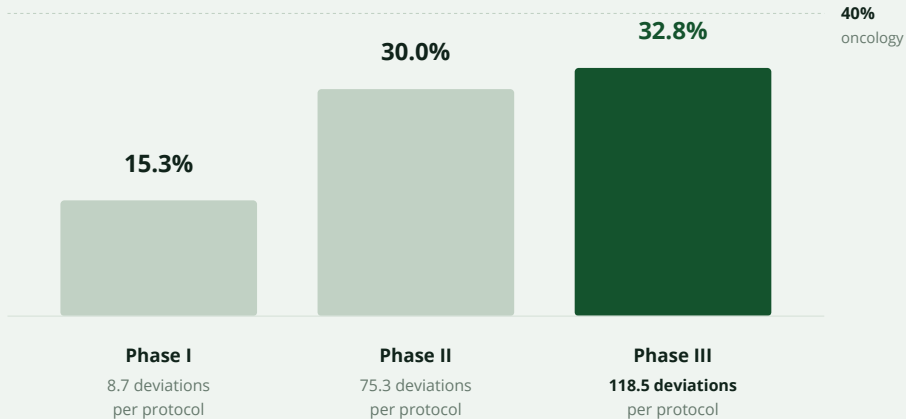
Medication reconciliation happens at the visit, so the record is whatever the participant remembers.

Nothing above reaches the site any other way. A course that ended six weeks ago is the easiest thing in the world to forget.

Reconciliation times: AHRQ *Patient Safety and Quality*; time-and-motion study of 103 reconciliations, *BMC Health Serv Res* 13:485. Hospital settings.

By Phase III, one participant in three has a protocol deviation

187 protocols from 20 sponsors and CROs — Tufts CSDD, 2022

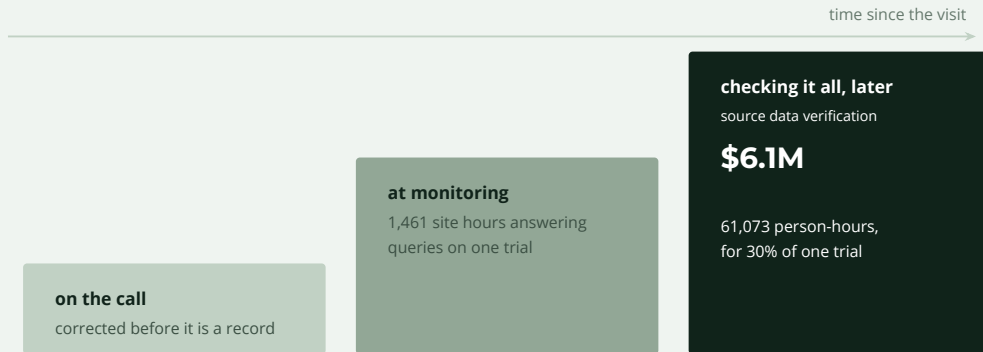


bars: share of enrolled patients with at least one protocol deviation

Getz et al., *Therapeutic Innovation & Regulatory Science* 56:632–636, 2022. All deviation types, not medication-specific.

The same error, found later, costs \$4,862

Retrospective verification of one phase 2 trial — I-SPY COVID, 2025



All of that verification found errors in 0.36% of the fields it checked.

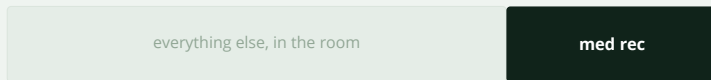
Checking everything late is the expensive way to learn almost nothing. Catching it on the call is the cheap way.

Communications Medicine 5, 2025 · 333 of 1,111 patients, 10,101 of 44,486 eCRFs verified.

Ask days before, not minutes before

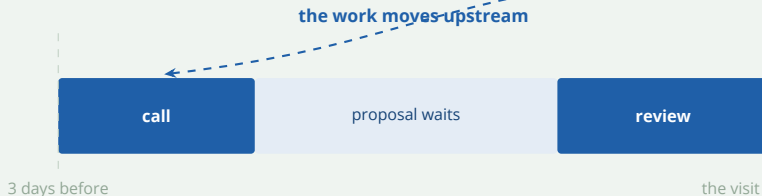
Today

at the visit



With ReconMed

three days earlier



the agent calls

identity, the log read back, side effects, and the protocol's own rules

nothing is written

proposals are staged; the medication log is untouched until a person signs

the coordinator reviews

a draft with its reasoning and every lookup attached

Three days is long enough to ask someone to go and read the label.

Before, and after

Before

the coordinator, by hand



The participant arrives

nothing collected until now



Go through the list together

in the room, from their memory



30-60 min



Enter it into the record

whatever was recalled, is the record



ReconMed

the record, live

● live



the record is already loaded



the call



Add



Modify



Stop



review, not interview

the work is already done

Asked days earlier, while there is still time to check a bottle.

Reconciliation duration: AHRQ, *Patient Safety and Quality*, per admission.

Calling participants between visits already works

25,829 pharmacist calls across 12 clinical trials — Martinez et al., 2017

45.5%

of those calls produced at least one intervention

Around 55% were rated high-impact — affecting trial outcomes or adherence.

Participants with an issue found by phone

adverse events



medication use against protocol



concurrent medications

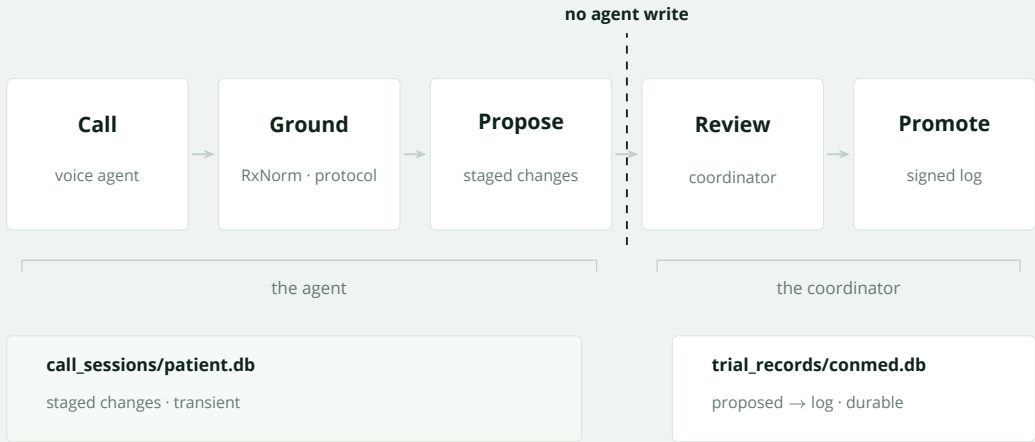


A pharmacist made every one of those calls.

That is why it is not standard practice. ReconMed makes the call, and a coordinator reviews only what it found.

Martinez et al., *Clinical Therapeutics* 39(3), 2017. 3,873 participants; 12 trials in type 2 diabetes.

The agent proposes. A coordinator decides.



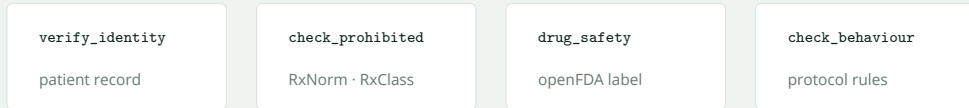
the medication log is written only on Promote, under an electronic signature

Two models, because a realtime agent cannot wait

The Talker speaks. The Thinker plans between turns and never touches the line.

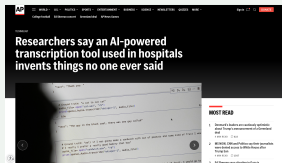


Every claim resolves to a lookup



Deepgram nova-3 in, aura-2 out · GPT-5.6 Luna both sides · prompts are files, swappable without a deploy

The model is not the source of truth

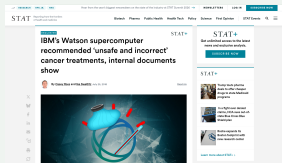


AP News

26 Oct 2024

Researchers say an AI-powered transcription tool used in hospitals invents things no one ever said

"hyperactivated antibiotics" — a drug that does not exist, invented in a patient's notes.

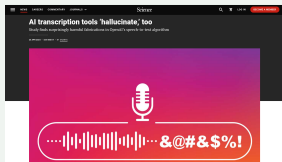


STAT

25 Jul 2018

IBM's Watson supercomputer recommended "unsafe and incorrect" cancer treatments, internal documents show

recommended a drug that could cause severe or fatal bleeding — for a patient already bleeding.



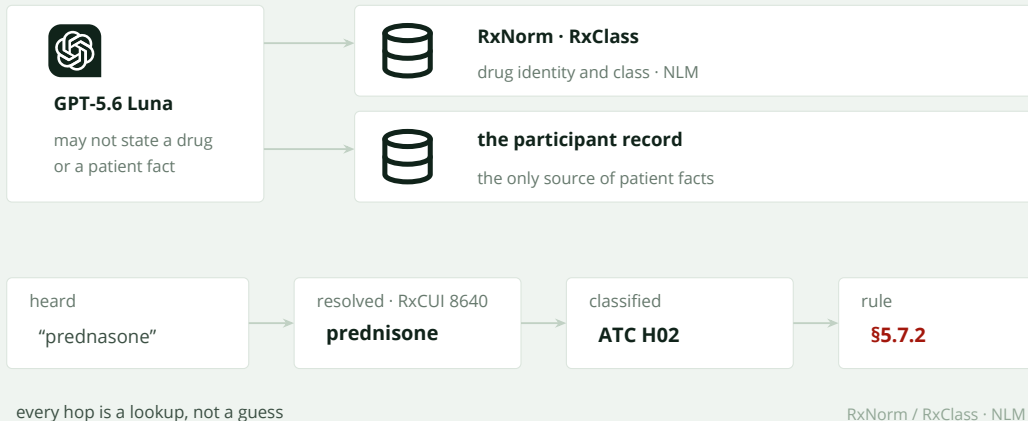
Science

26 Apr 2024

AI transcription tools "hallucinate", too

made up whole sentences in about 1.4% of the transcriptions tested.

The agent may not answer from memory



Review queue

6 proposed changes



Prohibited

Prednisone
20 mg daily · oral

Protocol §5.7.2



matched drug
prednisone

matched class
systemic corticosteroid
ATC H02

protocol section
§5.7.2
excluded above 10 mg/day

What is on file, and what the call proposed

Review queue

7 proposed changes · 2 prohibited

on file

from the call

Add

—



Atorvastatin 20 mg nightly



pending

Modify

Metformin 500 mg twice daily



Metformin 1000 mg twice daily



accepted

Stop

Ibuprofen 400 mg as needed



stopped — GI bleed reported



rejected

each row: confirm, edit, reject · and “why” opens the tool trace

Promote 1 accepted

The record a monitor will ask for

21 CFR Part 11: why it changed, who approved it, and when.

Audit trail

append-only

10:24	Voice agent reason: —	proposed stop ibuprofen
10:31	J. Okafor, CRC reason: GI bleed reported	accepted
10:31	J. Okafor, CRC reason: confirmed with participant	edited dose
10:32	J. Okafor, CRC reason: electronic signature applied	promoted 1 change

Electronic signature

J. Okafor, CRC

Clinical research coordinator

Approved

2026-09-19 10:32

A signature carries printed name,
timestamp and meaning (11.50).

Every entry carries who / when / why.

One machine, no network

Three real published protocols, 25 participants — trials real, participant data synthetic.

Everything local



conmed.db

the trial record



patient.db

the call in progress



drugs.db

RxNorm



general_health.db

plain-language guidance

delete any of them and rebuild from the source data

Or a server

`DATABASE_URL=`

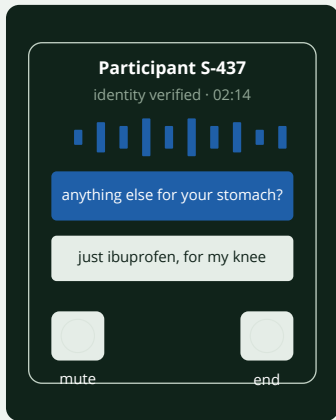
`postgres://...`

no database is opened by path:
one environment variable, same code

MIT · open source

the same schema, three methods

See it make the call



scan — the whole project

github.com/Samuel-O-M/HackMIT-2026

What it changes

14

calls run

33

changes proposed

11

confirmed unchanged

3

real protocols

coordinator

- starts from a draft, not a blank form

sponsor

- fewer deviations found late

participant

- nothing to remember or repeat

Open source (MIT) · point it at your own model · pilot it at one site

Sources

Evidence

Getz et al., *Ther Innov Regul Sci* 56:632, 2022

187 protocols · deviation rates by phase

Communications Medicine 5, 2025

I-SPY COVID · retrospective SDV cost

Martinez et al., *Clin Ther* 39(3), 2017

25,829 pharmacist calls · 12 trials

FDA final guidance, 2026

washout periods and concomitant medications

AP · STAT · Science, 2018–2024

reported failures of clinical AI

Trial and protocol data are real and public. Participant records are synthetic.

Grounding, live

RxNorm · RxClass

NLM · drug identity and class

openFDA drug label

FDA · documented adverse reactions

ClinicalTrials.gov

the three trials in this demo, verbatim

Published study protocols

prohibited medications, parsed from the PDF

CDISC SDTM · 21 CFR 11

record shape and signature



ReconMed

Automated Medicine Reconciliation

Thank you · questions

HackMIT 2026